

Coffee Shop Sales and Demand Analysis

Maven Analytics Dataset January–June 2023

Business Question:

How can the coffee shop chain use transaction data to understand sales growth, store performance, demand patterns and product mix, and identify practical opportunities for improving operations?

Data and Method:

This analysis uses 149,116 transaction records from the Maven Analytics Coffee Shop Sales dataset, covering 1 January to 30 June 2023. Revenue was calculated as transaction quantity multiplied by unit price. The data was checked for blanks, duplicate transaction IDs, and non-positive quantities or prices. Excel Tables, calculated columns, PivotTables, PivotCharts, slicers, and percentage comparisons were used to analyse monthly, store, weekday, hourly, and product performance.

Key Findings:

1. Revenue growth

Revenue increased from €81,677.74 in January to €166,485.88 in June, representing growth of approximately 103.8%. Average revenue per transaction remained almost unchanged, indicating that growth was driven mainly by an increase in transaction volume.

2. Morning demand concentration

Demand peaked at 10:00. Between 10:00 and 11:00, transactions fell from 18,545 to 9,766, a decline of 47.3%, while revenue decreased by 47.8%. This indicates a sharp transition after the morning peak (see Figure 1)

3. Product concentration

Coffee and Tea generated 38.63% and 28.11% of total revenue, respectively. Together, they produced 66.74% of all revenue. Barista Espresso was the highest-revenue product type at €91,406.20.

Recommendations

1. Morning operations: Review staffing, preparation, and product availability during the 8:00–10:00 peak. Where operationally possible, schedule breaks and restocking after the morning rush.

2. Product strategy: Protect the availability of high-revenue beverages and test bakery bundles or add-on offers to determine whether average transaction value can be increased.

Limitations

The dataset contains six months of transactions and does not identify unique customers. It does not include costs, profit margins, staffing, inventory availability, promotions, weather, or local events. Therefore, the analysis identifies patterns but cannot prove their causes. The source does not specify a currency; euro formatting was applied as a reporting assumption.

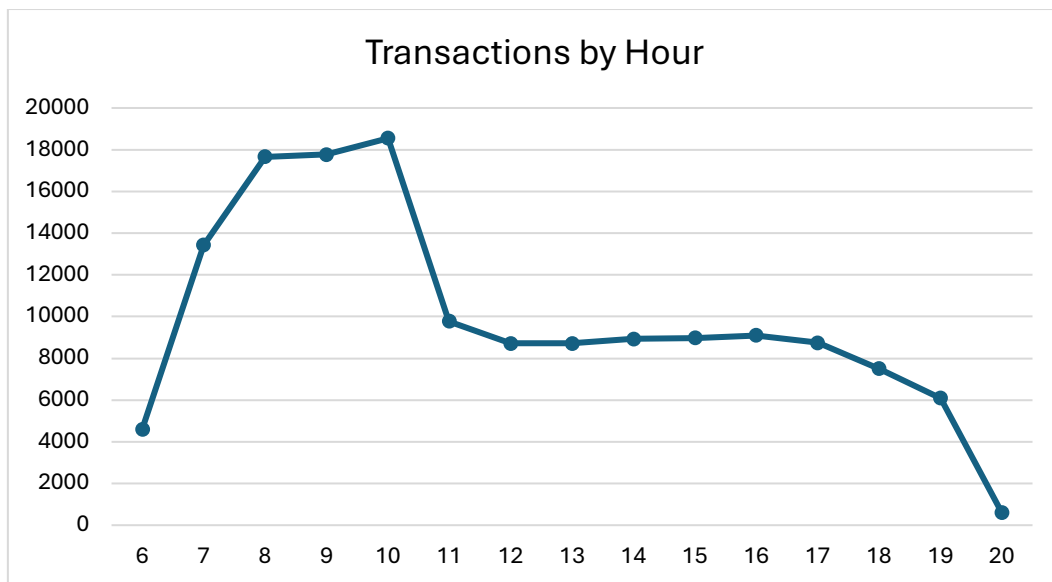


Figure 1. Hourly transaction volume, January-June 2023.

Source: Author's analysis of the Maven Analytics Coffee Shop Sales dataset.